

SEQUENTIAL CANDIDATE SELECTION FOR SCORE FOLLOWING IN SHEET MUSIC IMAGES

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ABSTRACT

Score following aligns a live performance to its score in real time, committing to a position at every instant with no lookahead. The two sides are different kinds of object, the performance audio and the score an image. The established response is to convert one into the other and align within a single modality, either by turning the page into symbols with optical music recognition or by turning the audio into notes with transcription, and each detour inherits the errors of its conversion. Sheet-image score following converts neither side and learns the correspondence directly, and its strongest systems condition a visual detector on the audio. What limits them is annotation, since performances aligned to notehead coordinates are costly enough that nearly all training data is synthesized, so these systems work on synthesized audio and lose much of their accuracy on a real instrument in a real room. Closing that gap is the field’s central open problem, and every remedy proposed for it has treated the gap as a failure of perception. We present evidence that it is not, since on real recordings the detector already proposes a position within the standard tolerance at every scored onset, and an oracle held to the same realtime commitments recovers almost everything the deployed rule loses. We therefore recast the task as sequential selection over the candidates the detector already emits, combining a parameter-free transition model with a set-based scorer that judges each candidate against its competitors in the same instant. This sets a new best on real piano recordings played straight through, holds across three released detectors that differ in heads and in training data, and shows that what the decision needs is its own temporal history rather than a better image representation. Where a performance leaves the written order, the same forward assumption that makes the decision tractable is what breaks, and we report that case rather than set it aside.

Index Terms— Score following, sheet music, audio-to-score alignment, realtime, music information retrieval

1. INTRODUCTION

Score following [1] is the online alignment of an acoustic signal to the score being performed. Given the audio observed up to frame t and a representation of the score, the system emits an estimate of the position the performer currently occupies, and it does so under a strict latency budget, in that the estimate for frame t must be produced before frame $t + 1$ arrives. The task supports automatic page turning [2], live computer accompaniment [3, 4, 5] and score-synchronised visualisation.

Aligning a performance to its score offline is a different problem, usually called audio-to-score alignment or music synchronisation. There the entire recording is available, a global method such as dynamic time warping settles every position at once [6], and even repeats and jumps can be resolved after the fact [7, 8]. Offline alignment underpins cross-modal retrieval and piece identification, where a recording is matched against a collection of scores [9]. Score following, also called music tracking [10], has none of these advantages. It observes only what has been played, commits to a position immediately, and never revises it. Everything below respects this constraint, including the oracles we use to measure what is achievable.

The remaining difficulty is that the two sides are different kinds of object. The performance is audio, the score is an image, and there is no shared space in which to compare them, a mismatch the literature answers in three ways.

1.1. Related work

One family converts the score, using optical music recognition to turn the page into symbols so that a classical follower can track the performance over a discrete set of score states. This line is well developed, spanning online dynamic time warping [6], duration-focused probabilistic models [11], hidden semi-Markov models that absorb performer errors and arbitrary repeats [12], particle filters for rests and tempo change [13], switching state-space models with hidden tempo [14], and multi-agent tracking for robustness to arbitrary deviation [10]. Its weakness is the conversion, since optical music recognition remains error-prone on real scores [15] and the follower inherits every mistake.

A second family converts the audio instead, transcribing the performance into notes and comparing symbol to symbol.

Recent work pairs realtime piano transcription with symbol-level tracking and reports strong accuracy [16], though the follower is then bounded by the transcriber, which restricts it to instruments transcription handles well.

The third family converts neither side and learns the correspondence between audio and pixels directly. Dorfer et al. [17] first classified position into staff-level buckets on image crops; reinforcement learning then framed tracking as an agent navigating a score strip [18, 19]; CUNet [20] moved to full pages, emitting an audio-conditioned segmentation whose centre of mass gives the position; and CYOLO [21] recast it as audio-conditioned object detection, with CYOLO-SB [22] adding heads for the enclosing system and bar. The concurrent CODA [23] cascades system, bar and note selection over the layout regions of the score, with beam search and learned transition penalties. It is the strongest published system on this benchmark, but it requires the system and bar bounding boxes of the page as an additional input at test time, so it presumes a layout annotation that the other methods, and ours, do not read.

Training any of these needs performances aligned to note-head coordinates on the page, an annotation costly enough that nearly all of it is synthesized. A score is rendered to MIDI and the MIDI to audio, and the alignment follows from the rendering at no additional cost. The result is systems that work on synthesized audio and lose much of their accuracy on a real instrument in a real room. CUNet [20] falls from 85.5% of onsets tracked within 0.5 s to 22.4%, CYOLO-SB [22] from 89.3% to 79.9%, and CODA [23] from 95.5% to 88.3%. Closing this gap is the field’s central open problem, and every remedy proposed for it has been a perception remedy, whether reverberant augmentation [21], extra proprietary recordings [22], or a new architecture [23].

1.2. Diagnosis

We tested that premise before accepting it, starting from the observation that a conditional detector does not produce a position directly but scores thousands of candidate noteheads and reports the most confident one. Keeping the 256 most confident, we find that on real room-microphone recordings a candidate within the 0.5 s tolerance is present at every one of 4,149 scored onsets. An oracle free to pick any of them is therefore perfect, but could not be run live. The one that matters must commit at each instant and never move backwards, exactly as a deployed follower must, and it still reaches 99.6%. The deployed rule reaches 80.0% (Fig. 1). Nineteen of the twenty points lost to real audio survive the realtime constraint and are recoverable from what the model already saw. At this tolerance, then, reverberation degrades the detector’s confidence ordering more than the visual evidence beneath it, and Sec. 4 qualifies how far that holds as the tolerance tightens.

We therefore recast sheet-image score following as a sequential decision over the positions the page affords, rather

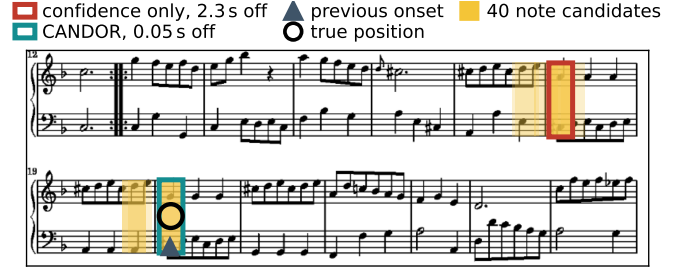


Fig. 1. One instant of a real room-microphone recording. In yellow, 40 of the candidate positions the detector scores. The most confident (red) lies on the wrong staff, 2.3 s from the truth; CANDOR (teal) takes a less confident candidate on the true note (ring). The correct position was available and was not chosen, which is the characteristic failure on real audio.

than the regression of a coordinate. CANDOR makes that decision with two parts, a transition model scoring a displacement against how much time has passed, and a set-based scorer judging each candidate against its competitors in the same instant, trained listwise on synthesized audio alone. The decision is causal and greedy, with no lookahead and no beam, and costs 1.25 ms per instant on a single CPU thread against the 50 ms available at 20 frames per second.

Our contribution is to relocate the synthetic-to-real gap in this task and to act on it. Candidate-set oracles place the gap in the decision rather than in perception, and CANDOR, a decision model built on that finding, sets a new best on real recordings by a margin that holds across three released detectors differing in heads and in training data.

2. METHOD

2.1. Backbone

Every sheet-image follower of the last five years shares one perception module, a convolutional network over the score page whose feature maps are modulated by the audio. The mechanism is feature-wise linear modulation [24], which injects an external vector into a network by scaling and shifting its activations,

$$f_{\text{FiLM}}(x) = s(z) \odot x + t(z), \quad (1)$$

where s and t are learned linear maps of a conditioning vector z and $\odot$ is elementwise. The page alone cannot say where the performer is, so z carries the recent audio and the visual features become a function of what has just been heard.

CYOLO [21] encodes the last two seconds of audio into z with a spectrogram network and places FiLM layers through the downscale and upscale blocks of a YOLO detector, which then predicts a bounding box for the notehead currently sounding. CYOLO-SB [22] keeps that arrangement and adds heads for the enclosing system and bar, so one 52×52 feature map yields all three. CODA [23] conditions its own backbone the

same way, at the deeper stages, before cascading selection over layout regions supplied with the score.

We take this module as given rather than as a design choice, and use the released CYOLO detectors to propose noteheads. No detector weight is updated anywhere in this work, and three released instances are used, differing in their heads and in their training data, so the conclusions concern the family and not one checkpoint.

2.2. CANDOR

CANDOR, for candidate ordering, sits after the backbone and replaces only its final step. At each instant the detector offers a few hundred possible notehead positions, and where it would report the most confident of them, CANDOR asks which one is most compatible with where its own previous decision was and with what the rest of the set looks like. Those two questions are the two parts below, and answering them requires carrying state from one instant to the next.

2.2.1. Candidate positions

A performance is processed in instants $t = 1, \dots, T$, and at each one the backbone reads the audio so far and the current page and scores note boxes $b_i = (x_i, y_i, w_i, h_i)$ with confidence o_i ; we keep the $K = 256$ most confident, $\mathcal{C}_t$. Staves are concatenated along one horizontal axis, the unrolled coordinate u_i , so positions across a line break are adjacent. The released detectors score 8,112 boxes per instant and report

$$\hat{i}_t = \arg \max_{i \in \mathcal{C}_t} o_i, \quad (2)$$

which is the degenerate selector. CANDOR reads $\mathcal{C}_t$ and its own previous choice $\hat{u}_{t-1}$, and nothing else.

2.2.2. Transition model

Selections must form a plausible trajectory, and between instants Δ_t frames apart the performer advances by a roughly predictable amount, so the displacement $d_i = u_i - \hat{u}_{t-1}$ is scored by a floored Gaussian whose mean grows with elapsed time,

$$\log \pi_i = \max \left(-\frac{(d_i - \mu \rho_t)^2}{2\sigma^2}, J \right), \quad (3)$$

with $\rho_t = \text{clip}(\Delta_t/\Delta_0, 0.2, 8)$ and $\Delta_0 = 5$ frames, the mean inter-onset gap in training. The floor J keeps a jump possible at bounded cost. Combined with confidence, $h_i = \log o_i + \log \pi_i$ already recovers a large part of the gap on its own.

2.2.3. Set-based scorer

The transition model treats candidates independently given the history, so what it cannot express is comparative, whether this box, among those on offer in this instant, looks like a real

notehead rather than a spurious response. This is what real audio in particular calls for, since reverberation smears the onset the conditioning vector is built from and the detector answers with many moderately confident boxes rather than one sharp response, so absolute confidence is poorly calibrated while the shape of the set is still informative. Each candidate is described by $\phi_i \in \mathbb{R}^{33}$ holding its confidence relative to the instant's best, box geometry, motion relative to the history, relation to any coarse regions the backbone emits, and a neighbourhood group describing how crowded and how typical the box is among its neighbours. We also read f_i , the 128-dimensional FiLM-conditioned feature from which the detection head scored the box, so the decision sees the representation perception used. A permutation-invariant network [25], as in learned re-scoring of detections [26], embeds the set and pools a context shared by all candidates,

$$e_i = g([\phi_i; \text{ReLU}(W f_i)]), \quad c = \left[\frac{1}{K} \sum_j e_j; \max_j e_j \right], \quad (4)$$

$$s_i = r([e_i; c]), \quad (5)$$

with g and r two-layer perceptrons (64, 32 and 64, 1) and $W \in \mathbb{R}^{64 \times 128}$. The maximum is taken per dimension, so every candidate is judged against its competitors.

2.2.4. Training and inference

The decision is supervised listwise, so with ε_i the error in frames of candidate i the target is $q_i \propto \exp(-\varepsilon_i/\tau)$, $\tau = 3$, and

$$\mathcal{L} = - \sum_{i \in \mathcal{C}_t} q_i \log \frac{\exp s_i}{\sum_j \exp s_j}, \quad (6)$$

so the scorer learns to order a set rather than to regress a coordinate. History features come from previous decisions, perturbed on 30% of samples by Laplace noise of scale 30 px to expose training to the drift of sequence prediction [27]. At inference CANDOR decides

$$\hat{i}_t = \arg \max_{i \in \mathcal{C}_t} \beta s_i + (1 - \beta) h_i, \quad (7)$$

causally and greedily, one pass over 256 candidates per instant, in 1.25 ms on a single CPU thread against the 50 ms available at 20 fps. Training on the residual $s_i - h_i$ and deciding $\arg \max_i (s_i + h_i)$ is Eq. 7 at $\beta = 0.5$ and eliminates all four constants.

3. EXPERIMENTAL SETUP

3.1. Dataset

All experiments use MSMD [9], which pairs public-domain LilyPond engravings with MIDI-derived alignments for solo piano. Its repertoire is Western classical, weighted towards the Baroque and Classical periods and dominated by Bach, with Beethoven, Mozart, Handel, Schumann, Chopin, Czerny

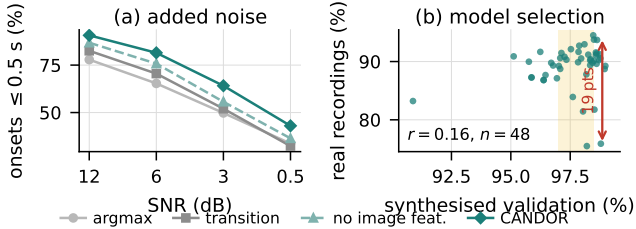


Fig. 2. (a) 80 held-out pieces under a measured room impulse response and additive noise, 40,594 onsets per level. (b) Each point is one of 48 trained decision models. Half of them fall in the shaded 1.5-point band of validation accuracy and still differ by 19 points on the real recordings.

and Burgmüller making up most of the remainder, alongside a smaller set of Romantic character pieces, traditional tunes and early ragtime. Textures range from two-voice inventions to dense Romantic writing, which matters here because the number of simultaneous noteheads sets how many plausible candidates a page affords at each instant.

The decision model is trained on 353 pieces, synthesized and convolved with measured room impulse responses [28], and validated on 19, using AdamW ($3 \cdot 10^{-3}$, cosine, 40 epochs, batch 256). The evaluation set is the real-performance benchmark of [19], 16 pieces of Bach, Schumann, Mozart, Chopin and Mussorgsky spanning 25 pages and 4,149 onsets, played on a Yamaha AvantGrand N2 and captured with a room microphone. A held-out set of 80 further pieces is rendered with a real impulse response and noise from 12 to 0.5 dB SNR.

We report the percentage of onsets whose predicted position maps to within 0.05 to 5 s of the true time, 0.5 s primary, with 95% intervals bootstrapped over pieces [29].

Three kinds of data enter at three separate stages and it is worth keeping them apart. The decision model is trained on synthesized audio alone and never sees a real recording. The four constants (μ , σ , J , β) are then calibrated on real recordings leave-one-piece-out, so for each evaluation piece they are chosen on the other 15 with pieces weighted equally and none influences its own decoding. Evaluation is on the piece held out. Leaving a piece out of its own constants does not change the fact that the benchmark as a whole participates in choosing them, which the published baselines did not require, so the residual form of Eq. 7 is carried alongside as the variant that fits nothing to any real recording. Seeds move results by about a point, so rows are seed means with deviations.

4. RESULTS

One clarification belongs before the table. CANDOR orders proposals, so it applies to a follower that produces a candidate set and then discards it. CODA [23] is not such a target: its final stage regresses a continuous bar-local coordinate rather than choosing among proposals, and the sets it does select

over, systems and bars, are supplied by layout annotation and already decoded with learned transition penalties and beam search. It appears below as the strongest published system on these recordings, not as a backbone CANDOR could be attached to.

Table 1 sets CANDOR against the published literature, taken from [23], on the same recordings, all of which are played straight through; Sec. 4.2 reports what happens when a performance is not. There Δ is the gain at 0.5 s over the perception model each row is built on, $\ddagger$ marks the one method reading the system and bar boxes of the page at test time, and $\dagger$ a checkpoint trained on proprietary recordings. CANDOR tracks 94.4% of onsets within 0.5 s, 95% CI over pieces [a, b], where the previous best was 88.3%, and leads at every threshold. The margin is not an artifact of one perception model. Retrained on whichever candidates each model emits, the decision gains 20.4, 10.7 and 7.9 points on three of them, and on the weakest, whose training data is entirely public, it passes the point at which the strongest one begins. A decision model does not need good proposals, only proposals containing the answer. On the 94 synthetic test pieces CANDOR reaches 95.5% where argmax reaches 89.3%, and on 80 held-out pieces under a real impulse response and noise it beats the transition model at every level from 12 to 0.5 dB (Fig. 2a).

The tightest thresholds qualify this, since on CYOLO-SB candidates CANDOR reaches 69.0% and 71.4% at 0.05 and 0.1 s, below CODA’s 72.1% and 74.3%, because a decision is only as precise as the boxes it chooses between, so the mechanism recovers which notehead is active rather than the position within it.

Table 2 separates the parts over the same candidates, adding Eq. 3 to Eq. 2 and then the scorer, with n counting decision-model seeds at a fixed detector. The transition model alone, with no learned parameters, accounts for roughly half the gain. The residual variant is the one to read for the strictest protocol, since it fits no constant to any real recording and is therefore comparable with the published rows on equal terms: it reaches 88.7% against 80.0% for the argmax it replaces, still above the previous best, and the distance from it to 94.4% is what calibrating four constants on the other 15 pieces is worth.

4.1. Ablation study

Giving the scorer the FiLM-conditioned backbone feature, DINOv2 patch features [30] on the same grid, or no image feature at all yields 90.6, 90.9 and 90.9, a spread well inside the 0.9-point seed deviation. On synthesized validation audio the same three differ by 3.4 points, so those features are learnable and useful there and worth nothing on real audio. We also rebuilt the perception model itself, retraining the detector from scratch for a fixed 24-hour budget with a different audio encoder. These are the lower block of Table 1, not comparable with the converged checkpoints above it and reported because the decision gains a similar margin whatever it is attached to.

Table 1. Onsets tracked on 16 real piano recordings, 4,149 onsets.

Method	Onsets tracked (%) within threshold (s)					Δ
	≤ 0.05	≤ 0.1	≤ 0.5	≤ 1.0	≤ 5.0	
MM-Loc [17]	36.4	40.6	58.5	61.1	73.5	
RL agent [19]	18.5	20.0	47.6	60.3	90.1	
CUNet [20]	11.3	12.5	22.4	26.6	44.3	
CYOLO [21]	56.3	58.1	71.2	74.9	91.9	
CYOLO-SB [22]	61.0	63.0	79.9	83.2	96.0	
CYOLO-SB+A [†] [22]	68.2	70.6	86.5	89.1	98.1	
CODA [‡] [23]	72.1	74.3	88.3	91.4	98.7	
CANDOR _{CYOLO}	72.6	74.7	91.6	95.0	99.1	+20.4
CANDOR _{CYOLO-SB}	69.0	71.4	90.6	94.2	98.2	+10.7
CANDOR _{CYOLO-SB+A[†]}	76.0	78.5	94.4	96.9	99.6	+7.9
<i>Detectors we trained, 24 h from random initialisation; only the audio tower differs</i>						
CANDOR _{CYOLO-LSTM} (control)	58.2	60.6	83.3	87.5	98.3	+11.6
CANDOR _{CYOLO-Mamba}	62.8	65.7	91.5	94.6	99.3	+19.0
CANDOR _{CYOLO} , audio tower as [23]	44.7	47.4	76.2	83.9	96.5	+17.3

Table 2. Components of CANDOR on the real recordings at 0.5 s.

	n	argmax	+transition	full
<i>Perception model</i>				
CYOLO [21]	3	71.3	72.5	91.6 $\pm$ 1.4
CYOLO-SB [22]	6	80.0	86.5	90.6 $\pm$ 0.9
CYOLO-SB+A [†]	3	86.5	90.8	94.4 $\pm$ 0.9
<i>Decision model, on CYOLO-SB candidates (80.0, 86.5)</i>				
residual, no fitted constants	3			88.7 $\pm$ 0.9
without temporal history	3			83.7 $\pm$ 0.4
<i>Image features f_i, on CYOLO-SB candidates</i>				
FiLM-conditioned backbone	6			90.6 $\pm$ 0.9
DINOv2 [30]	1			90.9
none	1			90.9

Table 3. Recovery from 41 spliced discontinuities.

Decision rule	break present		no break	
	rec@1	post	rec@1	post
argmax, Eq. 2	27	27	15	26
CANDOR	0	8	0	9
+ silence trigger	15	17	0	9
+ confidence trigger	10	26	2	23

A frozen MERT-95M tower [31] with only a projection trained reaches 46.2 and overfits 12.6-fold on the 353 training pieces. Replacing the whole audio tower with the two-layer causal Mamba of [23], trained as they train it, reaches 76.2 against 83.3 for an identically budgeted control, while keeping the convolutional front end and replacing only the recurrence, which we call CYOLO-Mamba, reaches 91.5. The recurrence therefore benefits from a state-space model and the convolutional front end is load-bearing, which is the encoder ablation absent from [23]. Both variants leave the visual tower untouched. None of this rescues the gap on its own: the strongest of these

detectors merely reaches what re-deciding over an already released one achieves, and all of them sit below the 94.4 of Table 1. Replacing the visual stem with frozen DINOv2-base patches on the same 52×52 grid, against a control retrained identically, gives ..

Removing the features computed against the previous decision costs 7.2 points, the largest single ablation here. Teacher forcing attributes two thirds of the remaining error to drift, since given the true previous position the same scorer reaches 98.0% against the causal oracle’s 99.6%, which locates the remaining difficulty in the sequence of decisions rather than in the image.

Across 48 trained models, accuracy on the synthesized validation split correlates $r = 0.16$ with accuracy on the real recordings (Fig. 2b); the best-validation model scores 89.3 on real audio, below one reading no image features at all. A reverberant held-out set correlates more strongly ($r = 0.80$) yet still selects poorly. Since the whole field selects models on synthesized audio, this is a caution beyond our own experiments, and it is why our constants are cross-validated on the target recordings instead.

4.2. Score discontinuities

Eq. 3 assumes the performance moves forward, which is why the decision is tractable and also why a repeat or a jump breaks it. Following the protocol of [23] we splice three discontinuities into each performance at random destinations, re-running perception on the spliced audio so its conditioning is genuinely wrong afterwards, and excluding jumps across a page boundary where state resets anyway, leaving 41. Table 3 decodes the CYOLO-SB+A candidates and reports rec@1, the percentage of breaks first tracked correctly within 1 s, and post, the percentage of onsets tracked within 1 s over the five seconds after one.

The table reports a systematic failure, in which after a jump CANDOR performs considerably worse than the rule it replaces, 8% against 27% of onsets tracked over the next five seconds, because the transition model holds the trajectory at a stale position. Suppressing it on detected silence helps where a break exists and not otherwise, as its design implies; triggering on the scorer’s own confidence needs no break, recovers most of the loss and reaches argmax without passing it; the same ordering holds over 220 jumps on the synthetic benchmark.

Whether such a trigger belongs in the deployed decoder turns on what it costs when nothing goes wrong, which the spliced takes cannot show. Run on the unspliced recordings with identical constants, the confidence trigger costs 1.0 point and the two-hypothesis variant 2.5, each consistently across three seeds and so outside the 0.4-point spread between them. The confidence trigger therefore buys roughly eighteen points after a discontinuity for one point of continuous accuracy. We report CANDOR without it, since the evaluation repertoire is played straight through, and note that the trade favours the trigger for any repertoire in which repeats are taken. What the trigger establishes is that the two mechanisms compose: CANDOR is a rule for selecting among candidates while the performance runs forward, discontinuity detection is a separate decision about when that assumption has lapsed, and the second can be placed in front of the first at a stated price. A transition model strong enough to be worth ten points on continuous playing is strong enough to be wrong at a discontinuity, and reconciling the two within one rule is open.

5. CONCLUSION

The synthetic-to-real gap in sheet-image score following has been read as a failure of perception and answered with better audio, more data and new architectures. The measurement reported here places it elsewhere. On real room-microphone recordings the detector already proposes a correct position at every scored onset, and an oracle held to the same realtime commitments recovers almost all of what the deployed rule loses. At the standard tolerance, then, most of what is needed to follow the performance survives the room, and what fails is the rule that reads a position out of it.

Acting on that, CANDOR orders the candidates the detector already emits and takes real piano recordings from 80.0% to 94.4% of onsets within 0.5 s, against 88.3% for the best published system on a repertoire played straight through, at 1.25 ms per instant on a single CPU thread. The margin holds across three released detectors differing in heads and in training data, which suggests the finding concerns the family of audio-conditioned detectors rather than one checkpoint, and it is obtained without retraining any of them and without layout annotation at test time.

Within this framework the decision depends far more on its own temporal history than on the image representations we tested, since three image representations including none at all

give the same real-audio accuracy while removing the history costs more than any other ablation we ran.

Comparing a decoder of CODA’s kind with ours over one shared candidate set is a worthwhile experiment and a different one. Two results temper what is claimed here. Accuracy on synthesized audio barely predicts accuracy on real audio across 48 trained models, which makes model selection on a synthetic proxy unreliable for the whole field and not only for us. And the forward-motion assumption that makes the decision tractable is what fails at a repeat, where CANDOR is worse than the rule it replaces and no recovery mechanism we tried improves on that rule. Reconciling continuous tracking with discontinuity remains open, and pitch agreement between a candidate and the audio is the natural next source of evidence independent of the history.

Limitations. The correct page is given, all performances are piano, and 16 evaluation pieces cannot resolve differences of a point.

6. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical study that uses only publicly available data; no ethical approval was required.

7. ACKNOWLEDGMENTS

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